SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA04 **COURSE NAME: Operating Systems**

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.* *(BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Concept Mapping

Code of Conduct:

I S. Sharvesh / 192525458 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
S. Sharvesh

Annexure A

Concept Mapping

Question 1 **10 Marks**

Concept Mapping: Operating System Components and Services

An Operating System provides several services to users and application programs.

Task:

Match each Operating System Component with the most appropriate Operating System Service and explain the relationship.

Components	Services
Process Manager	File Management
Memory Manager	Process Execution
File Manager	Memory Allocation
Device Manager	I/O Operations
Security Manager	Protection & Security

After mapping, explain how these components collectively improve overall system performance.

Question 2 **10 Marks**

Concept Mapping: System Calls and Operating System Services

An application performs the following operations:

- Creates a new process
- Reads a file
- Allocates memory
- Sends data through a pipe
- Terminates execution

Task:

Map each operation to the appropriate system call category.

Operations	Categories
Create Process	Communication
Read File	File Management
Allocate Memory	Process Control
Send Data	Memory Management
Exit Process	Process Control

Write one example system call corresponding to each mapped category.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (100%)
Q1						
Q2						
Total						

ANSWERS – ANNEXURE A: CONCEPT MAPPING

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ANSWER 1 – Concept Mapping: Operating System Components and Services (10 Marks)

An operating system manages computer resources and provides services to programs. This concept map shows the service handled by each component and how the components work together.

OPERATING SYSTEM

↓ consists of five major components; each component provides a service ↓

COMPONENT	PROVIDES SERVICE	FUNCTION / EXAMPLE
Process Manager	**Process Execution**	Creates, schedules and terminates processes; supports multitasking.
Memory Manager	**Memory Allocation**	Allocates and releases RAM; tracks memory use.
File Manager	**File Management**	Creates, reads, writes, organizes and deletes files.
Device Manager	**I/O Operations**	Coordinates input/output devices through device drivers.
Security Manager	**Protection & Security**	Uses authentication and permissions to restrict unauthorized access.

↓ components cooperate to improve ↓

CPU UTILIZATION • MEMORY EFFICIENCY • FILE ACCESS • I/O PERFORMANCE • RELIABILITY • SECURITY

Relationships and Explanation

- **Process Manager → Process Execution:** Schedules CPU time and manages process life cycles; for example, a browser and music player can run concurrently.
- **Memory Manager → Memory Allocation:** Provides RAM to programs and tracks memory use, supporting efficient allocation and process isolation.
- **File Manager → File Management:** Organizes files and supports creating, reading, writing and deleting them; saving a document is a common example.
- **Device Manager → I/O Operations:** Coordinates hardware through drivers; for example, it manages a request to print a document.
- **Security Manager → Protection and Security:** Uses authentication, authorization and permissions to protect files and resources from unauthorized access.

How the Components Collectively Improve System Performance

For example, when I open a document, the Process Manager runs the application, the Memory Manager gives it RAM, and the File Manager finds the saved file. The Device Manager handles input/output, while

the Security Manager checks access. Together, they help the document open properly while other programs keep running.

Conclusion: The five components provide connected services that support efficient process execution, memory allocation, file handling, hardware communication and system protection.

ANSWER 2 – Concept Mapping: System Calls and Operating System Services (10 Marks)

A system call is the way an application asks the operating system kernel to perform a task. This concept map connects each operation with its service category and one example system call.

APPLICATION PROGRAM

↓ requests OS services through ↓

SYSTEM CALL INTERFACE

↓ operation → category → example system call ↓

APPLICATION OPERATION	SYSTEM CALL CATEGORY	EXAMPLE SYSTEM CALL
Create Process	**Process Control**	fork()
Read File	**File Management**	read()
Allocate Memory	**Memory Management**	mmap()
Send Data Through a Pipe	**Communication**	write()
Exit Process	**Process Control**	exit()

↓ requests are handled by ↓

OPERATING SYSTEM KERNEL — performs requests and manages resources

Relationships and Explanation

- **Create Process → Process Control → fork():** fork() creates a child process from the calling process.
- **Read File → File Management → read():** read() transfers data from an open file descriptor into a memory buffer.
- **Allocate Memory → Memory Management → mmap():** mmap() creates a mapping in a process's virtual address space and can provide anonymous memory with suitable flags.
- **Send Data Through a Pipe → Communication → write():** write() sends bytes to the write end of a pipe so another process can receive them from the read end.
- **Exit Process → Process Control → exit():** exit() terminates the program after normal library cleanup; the OS reclaims process resources.

Relationship Between System Calls and OS Services

In simple words, a system call lets an application request a service from the kernel. Process-control calls manage processes, file-management calls work with files, memory-management calls handle memory, and communication calls help processes exchange data. The kernel handles these requests.

Conclusion: System calls allow applications to request operating system services in a controlled manner and support process management, file handling, memory allocation and inter-process communication.

RUBRIC ALIGNMENT

CRITERION / WEIGHT	HOW THE ANSWERS ADDRESS IT
Concept Identification (25%)	Identifies key components, services, operations

	and categories.
Linkages (25%)	Shows component-to-service and operation-to-category relationships.
Organization (20%)	Uses hierarchical concept maps, tables and logical flow.
Integration of Literature (20%)	Connects concepts with practical examples and standard POSIX/Linux system calls.
Clarity (10%)	Uses labeled maps, concise explanations and clear connections.